Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations

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Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable — yet investigations unfold under closure pressure that structurally favors premature convergence on a single explanation before alternatives have been meaningfully compared. This paper reframes this failure as an HCI problem: not of generating explanations, but of maintaining comparison under conditions that prematurely collapse it. Interface design actively shapes which explanations become salient and which collapse — making the conditions of investigation reasoning a tractable target for HCI intervention. I introduce two design contributions: *comparison preservation* as a design principle for AI-supported reasoning systems in regulated environments, and *AI as comparison enforcer* as an inversion of the dominant model of AI in investigation support. I introduce Premature Accountability Convergence (PAC) as a concept that locates this failure not in individual cognition but in the interaction between closure pressure, documentation requirements, and the institutional incentives that shape what explanations remain comparable. I report a two-study empirical program with $N = 20$ GMP-experienced participants. In Study 1, a timed reflection protocol at three timepoints (0, +5, +10 minutes) establishes that comparison is time-sensitive: 15 of 20 participants produced single-cause, person-centered explanations at the initial timepoint, but 18 of 20 broadened their causal accounts within ten minutes. Confidence ratings averaged 1 out of 5 across all participants at the initial timepoint regardless of experience level, indicating that early convergence reflects institutional pressure rather than epistemic certainty. Post-task discussion revealed that all 20 participants selected explanations oriented toward documentation acceptability rather than investigative completeness. In Study 2, an AI-supported RCA Reasoning Scaffolder organized investigation into five stages enforcing explicit comparison before closure, with an AI component integrated into Stage 3 that detected convergence signals and triggered reopening prompts — functioning as a comparison enforcer rather than an explanation source. 18 of 20 participants produced broader causal accounts than in Study 1; 15 of 20 reconsidered their initial attribution, with 12 of those 15 shifting from person-centered to systemic or procedural explanations. All 20 participants expressed adoption intent while simultaneously identifying organizational transparency as a structural barrier to deployment. The paper makes three contributions: it introduces comparison preservation as a design problem in regulated investigation workflows, identifies PAC as a structural failure mode of accountability systems, and demonstrates that maintaining comparison is an interaction condition that can be reshaped through design.

Additional Key Words and Phrases: premature accountability convergence, root cause analysis, comparison preservation, AI scaffolding, GMP, closure pressure, regulated environments, human-AI interaction

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1 Introduction

Root cause analysis (RCA) in pharmaceutical manufacturing environments is expected to produce causal explanations that are both accurate and accountable. An investigation into a deviation should, in principle, compare multiple plausible explanations before arriving at a documented conclusion. In practice, investigations unfold under closure pressure — deadlines, documentation requirements, and procedural expectations that define when an investigation must be considered complete. Under these conditions, reasoning does not fail to generate alternatives; it fails to maintain them in a comparable state long enough for systematic evaluation. Convergence on a single explanation occurs not because other explanations have been eliminated, but because comparison between them has collapsed.

This paper reframes this dynamic as an interaction problem. The issue is not that investigators fail to generate alternatives, but that the workflow does not sustain those alternatives in a comparable state long enough for meaningful evaluation. Comparison collapses under conditions that prioritize closure, and the first viable explanation stabilizes as the documented account. Interface design determines which explanations remain visible and which are suppressed before comparison is complete — establishing the conditions of investigation reasoning as a domain in which HCI intervention is both possible and necessary.

I conceptualize this dynamic as Premature Accountability Convergence (PAC): the early stabilization of causal accounts that occurs when the comparison between alternative explanations collapses before reasoning has fully developed. PAC is not treated here as a cognitive bias or individual reasoning failure. It is treated as a structural outcome — a property of how investigative workflows organize, interrupt, and foreclose comparison under institutional pressure. This framing shifts the design question from "how do we help investigators think better?" to "how do we design workflows that keep comparison open longer?"

I report a two-study empirical program in pharmaceutical quality systems. Study 1 establishes an empirical baseline: using a timed reflection protocol with $N = 20$ GMP-experienced participants, I demonstrate that comparison is time-sensitive and that premature convergence reflects constrained investigative conditions rather than fixed cognitive limitation. Study 2 introduces an AI-supported RCA Reasoning Scaffolder that inverts the dominant model of AI in investigation support: rather than generating candidate explanations that investigators evaluate, the AI component detects convergence signals and requires their reopening — functioning as a comparison enforcer rather than an explanation source.

The contributions of this work are threefold. First, I introduce comparison preservation as a design principle for AI-supported reasoning systems in regulated environments — shifting the design question from reasoning support to structural intervention against premature closure. Second, I introduce AI as comparison enforcer as a design principle that inverts the dominant model of AI in investigation support. Third, I demonstrate empirically that comparison collapse is time-sensitive and therefore addressable through design, and that structured scaffolding can delay convergence and broaden causal accounts.

2 Related Work

Causal Explanation as Organizational Outcome

The foundational premise of this study — that causal explanations in regulated investigations are shaped by the organizational conditions under which they are produced — draws on Vaughan's (1996) account of the Challenger launch decision. Vaughan shows that the engineers and managers at NASA did not reach a catastrophic decision through negligence or bad faith; they reached it through a series of incremental, locally reasonable judgments that were

structured by production pressure, communication patterns, and the normalization of prior deviations. Applied to GMP deviation investigations, this framework implies that the causal explanations produced by RCA reflect the conditions of deliberation, not only the evidence available. PAC names the specific mechanism through which this shaping occurs: comparison collapses, and the first viable explanation becomes the documented one.

Recognition-Primed Decision Making and the Satisficing Problem

Klein's (1998) account of recognition-primed decision making (RPD) provides the cognitive background against which PAC becomes legible as a design problem. Klein demonstrates that expert decision makers under time pressure do not compare options — they recognize a situation as familiar, generate a single candidate course of action, and evaluate it for workability before acting. In regulated investigation contexts, the same expertise that enables rapid pattern recognition also enables rapid convergence, and the accountability stakes of investigation make that convergence consequential. Expertise, in this context, accelerates the very dynamic that investigation quality requires resisting: the more experienced the investigator, the faster and more confident the initial convergence.

Sensemaking and the Structure of Investigative Reasoning

Klein et al. [5] extend the RPD account through a sensemaking framework that treats diagnosis not as search through a fixed option space but as the progressive construction of a plausible account from incomplete evidence. Sensemaking is inherently comparative — explanations gain credibility not in isolation but against alternatives — yet the conditions of regulated investigation systematically suppress this comparative structure. PAC is, in these terms, a premature termination of sensemaking: the account that gets documented is the one that was plausible at the moment comparison collapsed.

Situated Action and the Conditions of Reasoning

Suchman's (1987) account of situated action establishes that reasoning is not plan-driven but shaped by the immediate conditions of its execution. An investigation workflow that requires a documented root cause before alternatives have been meaningfully compared creates conditions that structurally favor premature convergence, independent of individual reasoning quality. The scaffold developed in Study 2 is, in these terms, a redesign of the situated conditions of reasoning — an intervention in the environment rather than in the investigator.

Invisible Work and the Accountability Stakes of Comparison

Star and Strauss [8] show that work performed in the gaps of formal systems tends to become invisible. In regulated investigation workflows, the comparative work of holding multiple causal explanations in parallel is precisely this kind of invisible work: not required by the documentation template, not rewarded by closure expectations, and not visible in the final investigation record. The scaffold in this study makes this invisible comparative work structurally required, and therefore visible as a design concern.

Human-AI Interaction and Comparison-Preserving Design

The AI component of the Study 2 scaffold is designed around a principle that Amershi et al. [1] formalize: AI systems should support human control, make their capabilities and limitations clear, and avoid producing outputs that pre-empt or substitute for human judgment. In regulated investigation contexts, any system that generates causal conclusions or influences the direction of reasoning toward a specific explanation risks introducing a compliance risk. The AI

component in this scaffold is therefore constrained to a specific and narrow function: detecting signals of premature convergence and triggering prompts that reopen the comparison space.

Buçinca et al. [2] demonstrate that cognitive forcing functions, which require users to engage more deliberately with AI outputs before accepting them, reduce over-reliance without sacrificing efficiency. The PAC detection mechanism in Study 2 instantiates this principle in an investigation context, functioning as a forced pause rather than a recommendation engine — but extends it to a context in which the risk is not over-reliance on AI outputs but premature reliance on the investigator's own first account.

Prototype-Based Evaluation

The methodological approach builds on established HCI practice for prototype-based evaluation. Nielsen [7] demonstrates that think-aloud evaluation with small participant samples reliably surfaces the usability and reasoning support properties of an interface. Hutchinson et al. [4] establish the logic of using a prototype simultaneously as a design intervention and as an analytic probe: the technology probe makes visible the conditions it is designed to address.

AI-Assisted Decision Making and Interface Effects

Within HCI, work on AI-assisted decision making has examined how interface design shapes the quality of human judgment under uncertainty. Wang and Yin [11] demonstrate that explanation interfaces alter how decision makers engage with AI outputs, with effects contingent on domain expertise — findings that establish investigation interfaces as active shapers of reasoning rather than neutral conduits. Green and Chen [3] show that algorithm-in-the-loop settings systematically fail to produce reliable decision making even when users are instructed to exercise independent judgment, establishing that the structural conditions of human-algorithm interaction — not individual reasoning quality — determine whether comparison is meaningfully completed. These findings ground the design logic of comparison preservation: if interface structure shapes what explanations become salient, then redesigning that structure is a tractable intervention against premature convergence. Prior work has not examined premature causal convergence as a workflow design problem amenable to HCI intervention, nor has it introduced comparison preservation as a design principle for AI-supported reasoning in regulated investigation environments. This study addresses both gaps.

3 Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. Deviation investigations in these environments are formal quality events: they require documented root cause analysis, corrective and preventive action planning, and approval by qualified reviewers before closure. These requirements impose closure pressure — temporal and procedural expectations that define when an investigation must be considered complete — that shapes how reasoning develops before alternatives have been fully compared.

Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team. All participants had direct experience with GMP deviation investigation workflows, including root cause analysis, CAPA documentation, and quality review processes. Experience levels ranged from entry-level to senior practitioners with five or more years in GMP quality roles. This sample size is consistent with established think-aloud methodology,

in which $N = 20$ expert participants reliably surfaces the dominant reasoning patterns and usability concerns in a domain-specific population [7]. The same-sample design was a deliberate methodological choice to eliminate between-participant variance in domain knowledge as a confound, prioritizing internal consistency over sample independence. This design choice introduces the possibility of carry-over effects, which the study design addresses through three structural features. First, Study 2 was conducted after a minimum two-week interval following Study 1, reducing the salience of specific scenario details. Second, the scaffold in Study 2 structurally required participants to generate and evaluate hypotheses they had not previously committed to, making prior exposure to the scenario a less plausible explanation for broadened causal accounts than the scaffold's enforced comparison structure. Third, the primary outcome of interest — the shift from person-centered to systemic attribution — is directionally opposite to what carry-over alone would predict, as prior exposure would more likely reinforce rather than reverse initial convergence patterns.

Investigation Scenario

Both studies used the same standardized deviation scenario describing an out-of-specification result in an analytical testing context involving multiple instrument dependencies and procedural steps. The scenario was designed to support multiple plausible causal interpretations — including person-centered, procedural, equipment-related, and organizational explanations — so that no single explanation was obviously correct, and premature convergence on a person-centered cause was a realistic and common response.

Study 1: Timed Reflection Protocol

Study 1 was designed to establish an empirical baseline for how reasoning develops under closure pressure without structured support. Participants were asked to produce a causal explanation suitable for documentation within a fixed timeframe, using a standard RCA template consisting of a cause statement field, contributing factors section, and confidence rating. Reasoning was observed at three timepoints: an initial explanation produced immediately after reading the scenario (0 minutes), a revised or elaborated explanation after five minutes of unstructured reflection (+5 minutes), and a final explanation after ten minutes (+10 minutes). At each timepoint, participants recorded their current causal account and rated their confidence in it on a 1–5 scale (1 = very low, 5 = very high). Responses were collected via structured questionnaire at each timepoint. Participants recorded their current causal explanation in free text, selected a causal type from predefined categories (person-centered, procedural, equipment-related, organizational, or mixed), and rated their confidence on a 1–5 scale. Causal breadth was operationalized as the number of distinct causal factors participants identified in their written response and confirmed via their category selections. Causal type classification was performed directly by participants via structured response options in the data collection instrument, eliminating researcher coding and the associated inter-rater reliability requirements. This design choice reflects a deliberate alignment between the construct being measured — investigators' own causal framing — and the measurement method. This approach also addresses a potential concern about self-report bias: investigators in GMP environments are trained to categorize causal types as part of their routine documentation practice, making participant-led classification a professionally grounded judgment rather than a naive self-assessment.

Study 2: RCA Reasoning Scaffolder

Study 2 introduced an AI-supported RCA Reasoning Scaffolder designed to preserve comparison between alternative explanations by structuring the investigation workflow around explicit comparison steps before closure. The scaffold was implemented as a Streamlit web application and organized reasoning into five sequential stages (Figures 1–5):

(1) **Intake** (Figure 1): structured capture of deviation context without commitment to a causal conclusion. Hypotheses are listed and tracked as a live hypothesis panel showing active, narrowed, and dropped counts alongside a closure signal indicator.
(2) **Hypotheses** (Figure 2): generation of multiple causal candidates across person, procedure, equipment, environment, and organizational categories. Each hypothesis is assigned a PAC risk rating (High/Low) based on convergence-oriented language detection. Investigators are instructed not to narrow yet.
(3) **PAC Analysis** (Figure 3): rule-based detection of convergence-oriented language, with prompts to surface underexplored contributors and reopen alternative pathways. An AI component activates when individual-blame language is detected, surfacing alternative causal pathways (procedural/documentation, contextual/workflow, documentation/recording) and generating reopening questions.
(4) **Reasoning** (Figure 4): explicit per-hypothesis evaluation of contributing factors and evidence, with plausibility rating and investigation note for each hypothesis before any single explanation is selected.
(5) **Pre-Closure** (Figure 5): a required check confirming that alternatives were considered before final attribution, displaying which hypotheses remain active, narrowed, or dropped, and posing three verification questions before closure is permitted.

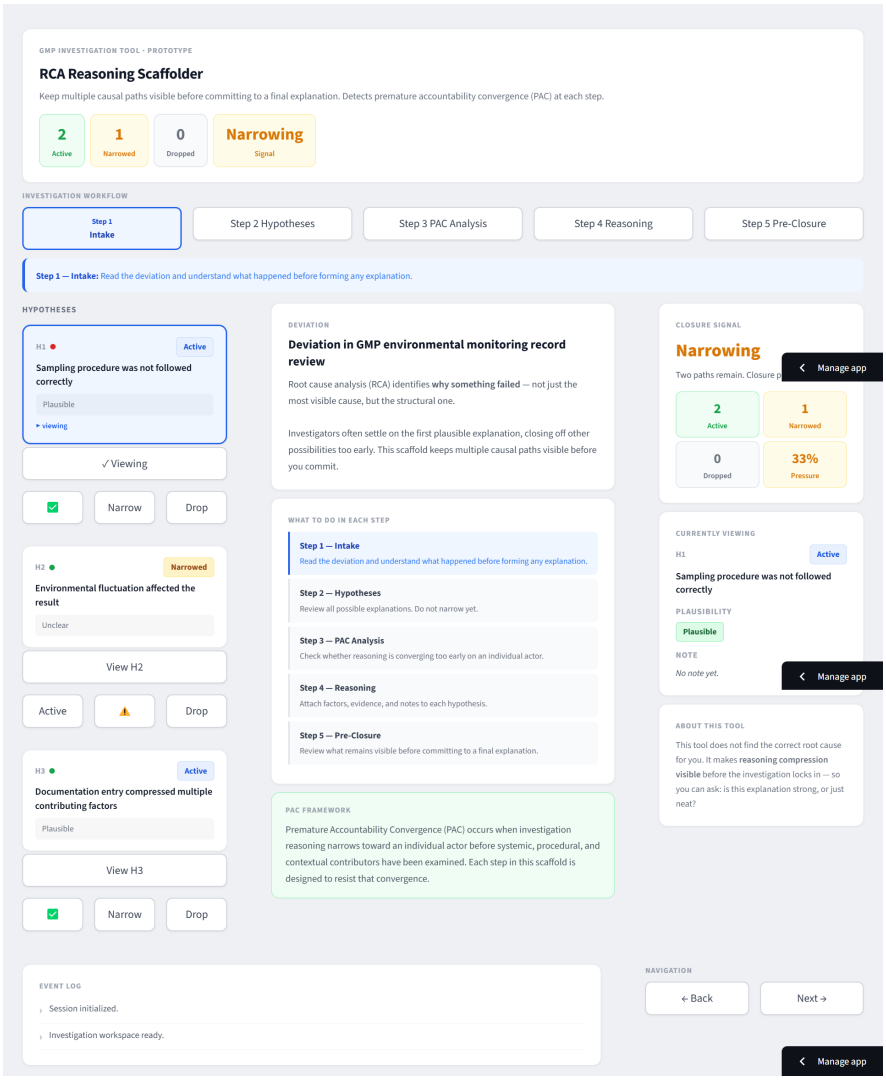


Fig. 1. RCA Reasoning Scaffolder: Step 1 (Intake). The hypothesis panel (left) shows three active hypotheses with current status. The central panel presents the deviation scenario and a step-by-step workflow overview. The closure signal panel (right) shows the current state: 2 active, 1 narrowed, 0 dropped, 33% pressure — indicating narrowing has begun before hypothesis expansion is complete.

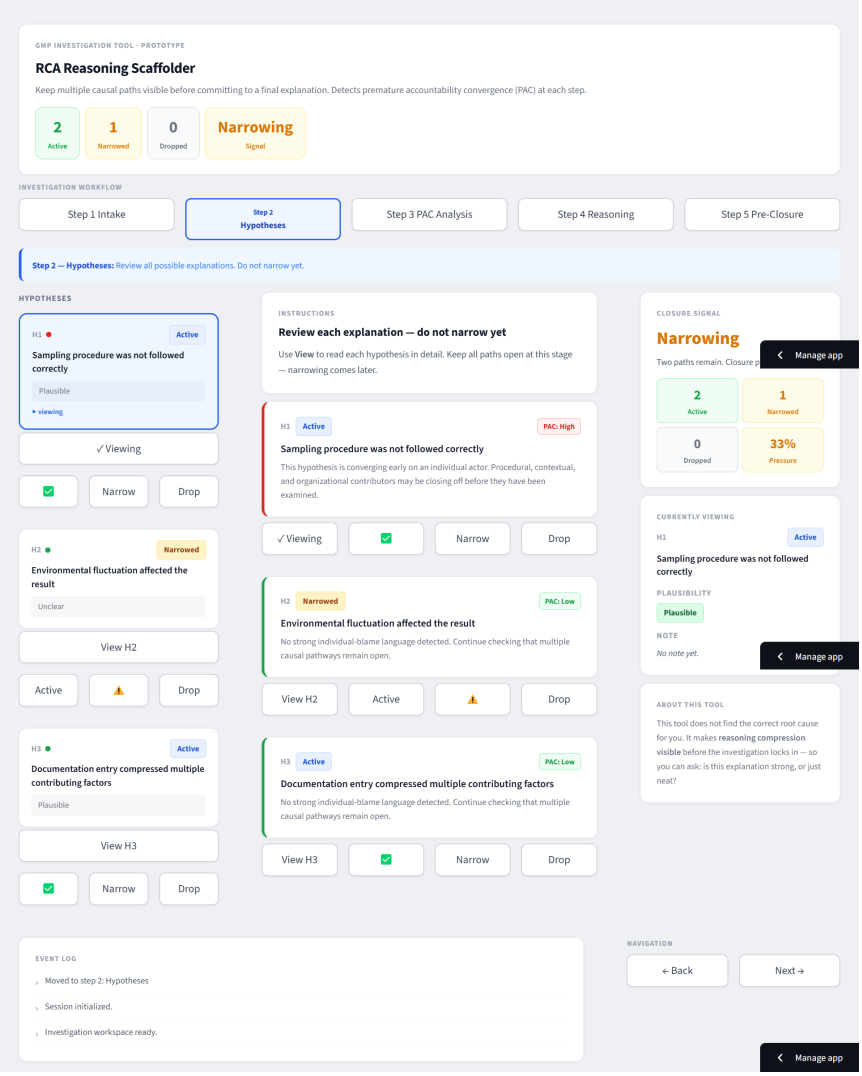


Fig. 2. RCA Reasoning Scaffolder: Step 2 (Hypotheses). Each hypothesis is displayed with its PAC risk rating: H1 ("Sampling procedure was not followed correctly") is flagged PAC: High, indicating convergence-oriented language targeting an individual actor; H2 and H3 are rated PAC: Low. The instruction "Review each explanation — do not narrow yet" enforces the comparison preservation principle at the stage level.

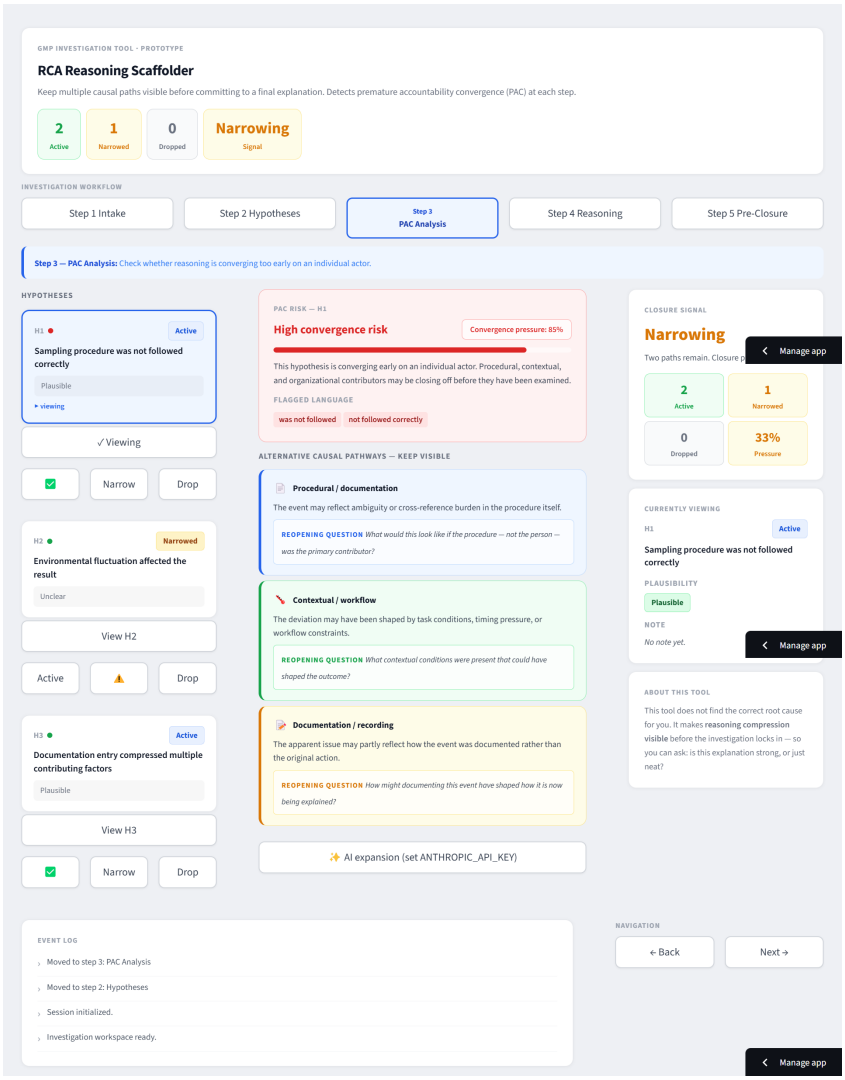


Fig. 3. RCA Reasoning Scaffolder: Step 3 (PAC Analysis) with AI layer active. H1 is flagged as high convergence risk (convergence pressure: 85%) with individual-blame language tokens highlighted ("was not followed," "not followed correctly"). The AI component surfaces three alternative causal pathways — procedural/documentation, contextual/workflow, and documentation/recording — each with a reopening question that must be considered before closure.

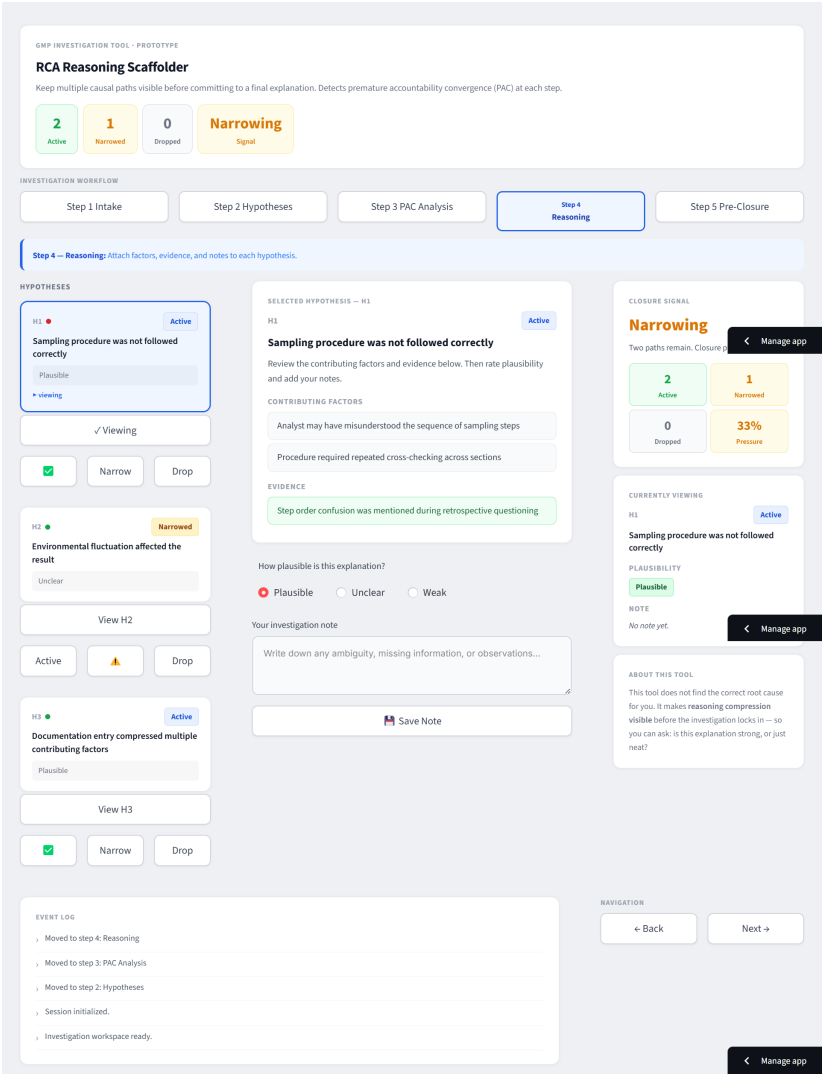


Fig. 4. RCA Reasoning Scaffolder: Step 4 (Reasoning). For the selected hypothesis H1, contributing factors and evidence are displayed for explicit evaluation. The investigator rates plausibility (Plausible/Unclear/Weak) and adds investigation notes before moving to the next hypothesis.

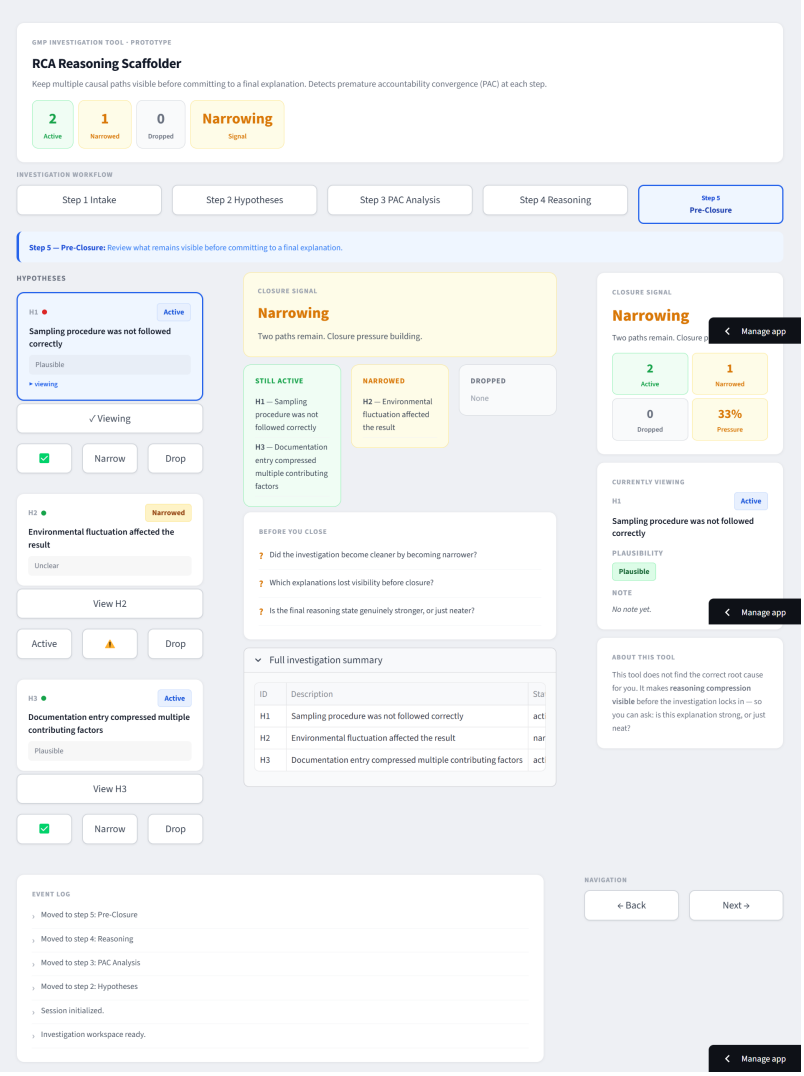


Fig. 5. RCA Reasoning Scaffolder: Step 5 (Pre-Closure). The closure signal panel shows the final hypothesis state: H1 and H3 still active, H2 narrowed, none dropped. Three verification questions are posed before closure is permitted: “Did the investigation become cleaner by becoming narrower?”, “Which explanations lost visibility before closure?”, and “Is the final reasoning state genuinely stronger, or just neater?”

The AI component was integrated into Stage 3 only. When the system detected signals of premature convergence — single-cause fixation, convergence-oriented language, or movement toward closure before alternatives had been articulated — it surfaced alternative causal pathways and generated reopening questions. The AI component did not suggest causes, evaluate hypotheses, or generate conclusions; it functioned exclusively as a conditional trigger for reopening the comparison space. This design reflects a principled inversion of the dominant model for AI in RCA support: rather than accelerating convergence by providing an anchoring explanation, the AI component acts against premature convergence by expanding the interpretive space when individual-blame language is detected.

Measures

Post-task assessment included five construct scales administered via structured questionnaire: usability (5 items; e.g., "This tool was easy to use"), reasoning expansion (5 items; e.g., "The tool helped me consider multiple possible causes"), PAC mitigation (4 items; e.g., "The tool prevented me from settling on a single cause too quickly"), perceived usefulness (3 items; e.g., "This tool would be useful in real-world RCA processes"), and blame pressure (1 item; "The tool reduced my tendency to focus on individual blame"). All Likert items used a 5-point scale (1 = strongly disagree, 5 = strongly agree). Confidence change was assessed via a pre/post 0–10 scale. Two behavioral items confirmed whether participants revised their initial explanation and considered more than one possible cause. Open-ended responses probed qualitative experience with the scaffold.

4 Findings

Study 1: The Time-Sensitivity of Comparison

Convergence is Fast. At the initial timepoint, 15 of 20 participants produced single-cause explanations — typically person-centered accounts attributing the deviation to analyst error, training inadequacy, or procedural non-compliance. Confidence ratings at the initial timepoint averaged 1 out of 5 across all participants, regardless of experience level — indicating that early convergence does not reflect genuine epistemic certainty but rather the institutional pressure to produce a documented explanation quickly.

With five additional minutes of unstructured reflection, 8 of 20 participants revised their accounts substantially. By ten minutes, 18 of 20 participants had either broadened their explanation or become more explicitly uncertain about the initial cause. This pattern establishes a core finding: comparison is time-sensitive. The collapse of comparison that produces PAC is not a fixed property of how investigators reason; it is a property of how much time is available for reasoning to develop.

The Selection of Documentable Explanations. Post-task discussion revealed a finding more consequential than the timepoint data alone. All 20 participants reflected that the explanation they select in practice is not simply the most plausible one but the most documentable one. As one participant described: "in a real investigation, you pick the explanation that closes the file. A systemic cause just opens more questions." Another noted: "I knew there were other factors. But if I write them all down, I own them." These accounts position premature convergence not as a reasoning failure but as a rational strategy: selecting a contained explanation is a structurally rational response to an accountability environment that rewards closure over completeness.

Study 2: Scaffold Effects

Causal Breadth and Delayed Closure. 18 of 20 participants produced broader causal accounts with the scaffold than in Study 1. The five-stage structure required participants to generate multiple hypotheses before committing to any, and the PAC detection layer triggered reopening prompts in sessions where convergence toward a single explanation had begun to occur. 15 of 20 participants reconsidered their initial attribution: of these, 12 shifted from person-centered accounts to systemic or procedural explanations. Post-task assessment across five construct scales confirmed the scaffold's effects. Usability averaged 5.0 out of 5, indicating that the five-stage structure presented no meaningful interactional barrier. Reasoning expansion averaged 4.8 (e.g., "The tool helped me consider multiple possible causes"; "I explored alternative explanations I would not have considered otherwise"), indicating that participants experienced a substantive broadening of their causal accounts. Perceived usefulness averaged 4.9, and PAC mitigation averaged

4.7 (e.g., "The tool prevented me from settling on a single cause too quickly"; "The tool encouraged me to consider system-level factors"). Blame pressure averaged 1.0 out of 5 — indicating that participants did not experience pressure toward individual attribution while using the scaffold.

Confidence ratings revealed a structurally important pattern. Participants entered the scaffold with relatively high confidence in their initial explanation — reflecting the same institutional pressure toward early closure documented in Study 1. Following scaffolded comparison, final confidence ratings were maintained or slightly decreased, while causal breadth increased substantially. This dissociation — sustained confidence alongside expanded reasoning — suggests that the scaffold did not destabilize participants' sense of epistemic security but redirected it: participants arrived at broader, more deliberated accounts without experiencing the scaffold as undermining their judgment. This pattern addresses a common concern about structured reasoning interventions, namely that requiring explicit comparison will produce uncertainty that practitioners find unworkable in high-accountability settings. The findings suggest the opposite: comparison-preserving scaffolding can expand causal reasoning while preserving the investigator's sense of epistemic grounding.

The usability ceiling observed across construct scales — including a mean of 5.0 out of 5 for usability — is better understood as a domain effect than a measurement artifact. GMP investigation workflows have no existing analogue for structured comparison scaffolding; participants evaluated the tool not against a competing system but against the absence of any structured support. The ceiling reflects the size of the gap the scaffold fills rather than the insensitivity of the measure.

Usability and the Conceptual Barrier. Participants did not struggle with the scaffold's five-stage structure and found the PAC detection prompts useful for identifying when they had been moving toward premature closure. The adoption barrier was not usability but conceptual: the idea that investigation reasoning deserves structured support — that comparison should be treated as a required step rather than an optional one — was initially unfamiliar. As one participant put it: "I never thought of the investigation as having a design. I just thought some investigators were more thorough than others."

The Organizational Adoption Problem. All 20 participants expressed intent to use the scaffold if available. Yet all 20 simultaneously identified a structural barrier to deployment: whether GMP organizations would adopt a system that makes investigation reasoning more transparent. The concern was specific to the audit trail implications of the scaffold's transparency. When the comparison process is documented, it produces a record of the alternatives that were considered and not chosen — and in a regulatory environment where that record is subject to audit, choosing not to pursue a systemic explanation becomes an accountable decision rather than an invisible one. As one participant noted: "the problem isn't the tool. The problem is that the tool would show that we looked at the systemic cause and decided not to pursue it. That's a conversation no one wants to have."

5 Discussion

Comparison Preservation as Interaction Design

The central design contribution of this study is the introduction of comparison preservation as a principle for AI-supported reasoning systems in regulated environments. More specifically, it reframes reasoning support as an interaction design problem: the issue is not only whether investigators can generate multiple explanations, but whether the workflow sustains those explanations in a comparable state long enough for meaningful evaluation. Comparison

preservation is therefore not a cognitive aid but a structural intervention. It maintains the availability of alternative explanations as an interaction condition.

Prior HCI work on AI decision support has established that interface structure actively shapes human judgment: explanation interfaces alter which features decision makers attend to [11], and algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible [3]. Work in this tradition has also focused on preventing over-reliance on AI outputs [2] and preserving human judgment [1]. Comparison preservation extends this work into a more specific and demanding context: in regulated investigation environments, the risk is not over-reliance on AI outputs but over-reliance on the investigator's own first account. The design challenge is not to keep AI out of the decision, but to structure the conditions under which the decision is made so that comparison has been genuinely completed before closure.

AI as Comparison Enforcer

The design contribution of this study is the introduction of AI as a comparison enforcer rather than an explanation source. The dominant model for AI in investigation support treats AI as a producer of candidate explanations that investigators evaluate — a model that accelerates convergence by providing an anchoring explanation that narrows the comparison space. The scaffold inverts this logic entirely. The AI component's function is not to suggest what caused the deviation but to detect when comparison has collapsed and to require its reopening. It acts against the structural tendency toward premature convergence rather than reinforcing it.

This instantiates the judgment-preservation principle that Amershi et al. [1] formalize, but in a more specific and demanding form. In a high-accountability regulated context, preserving human judgment means designing AI to resist the institutional pressures that produce premature convergence — not merely to avoid automating the final decision.

PAC as a Structural Problem

The finding that all 20 participants selected explanations oriented toward documentation acceptability rather than investigative completeness is not evidence of poor reasoning. It is evidence of rational adaptation to an institutional environment that rewards closed investigations over thorough ones. This extends Vaughan's (1996) account of organizational decision-making: the explanation that gets documented is shaped not only by the evidence available but by the accountability environment in which the investigator operates. The confidence finding sharpens this claim: if investigators converge prematurely while rating their confidence at 1 out of 5, convergence cannot be attributed to epistemic certainty. It is produced by institutional structure, not by belief.

The Organizational Adoption Problem

The most significant finding of this study is not about the scaffold's effectiveness but about the conditions of its deployment. All 20 participants expressed adoption intent; all 20 simultaneously identified the audit trail as a structural barrier. This combination — universal intent and universal concern — is not a contradiction. It reflects the gap between individual preference for investigative completeness and organizational incentive structures that reward documentation efficiency over thoroughness.

The scaffold demonstrates that comparison can be preserved through design. Whether that preservation is organizationally sustainable depends on whether the accountability environment rewards investigative completeness — a question that interface design alone cannot answer.

Limitations and Future Work

Several limitations bound the interpretation of these findings. The use of N=20 participants drawn from a single GMP quality team reflects a structural constraint of the domain rather than an arbitrary methodological choice. Usability and reasoning support evaluation in regulated investigation contexts requires participants with direct experience in RCA workflows, CAPA documentation, and deviation investigation — competencies that are not transferable from adjacent populations. This constraint structurally limits the available recruitment pool within any single GMP organization, and N=20 represents the boundary of what domain-valid sampling permits. The use of the same participant pool across both studies introduces potential carry-over effects. The single synthetic scenario may not generalize to the full range of deviation types and severity levels encountered in live quality systems. The organizational adoption concern raised by participants was not empirically investigated and would require separate study to examine systematically.

Future work should examine PAC in live investigation settings, where closure pressure, documentation requirements, and institutional incentives operate at full strength. The organizational adoption question is particularly important: understanding whether and under what conditions quality organizations would deploy comparison-preserving scaffolding — and what regulatory or organizational changes would be required to make a transparent investigation record an asset rather than a liability — is a research question that goes beyond HCI and into the organizational sociology of regulated work.

6 Conclusion

This paper introduced Premature Accountability Convergence (PAC) as a structural failure mode in regulated investigation workflows, and reframed it as an interaction problem concerning the conditions under which comparison is sustained or collapsed. Two empirical studies establish this claim and demonstrate one design response to it.

Study 1 showed that comparison is time-sensitive: 15 of 20 participants produced single-cause, person-centered explanations at the initial timepoint, with confidence ratings averaging 1 out of 5 regardless of experience — indicating that early convergence reflects institutional pressure rather than epistemic certainty. Post-task discussion confirmed that all 20 selected explanations oriented toward documentation acceptability rather than investigative completeness. Study 2 demonstrated that an AI-supported RCA Reasoning Scaffolder — organized around five stages of structured reasoning with an AI component functioning as a convergence detector rather than an explanation source — produced broader causal accounts in 18 of 20 participants, with 15 of 20 reconsidering initial attribution and 12 of those 15 shifting from person-centered to systemic or procedural explanations. Usability ratings averaged 5.0 out of 5. All 20 participants expressed adoption intent while simultaneously identifying the audit trail implications of transparent investigation reasoning as a structural barrier to deployment.

Three contributions follow. First, comparison preservation is introduced as a design principle for AI-supported reasoning systems in regulated environments — shifting the design question from reasoning support to structural intervention against premature closure. Second, AI as comparison enforcer is proposed as a design principle that inverts the dominant model of AI in investigation support. Third, the organizational adoption problem is identified as a first-class research concern: the conditions under which comparison-preserving tools can be deployed in regulated environments are as much a question of organizational sociology as of interface design.

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